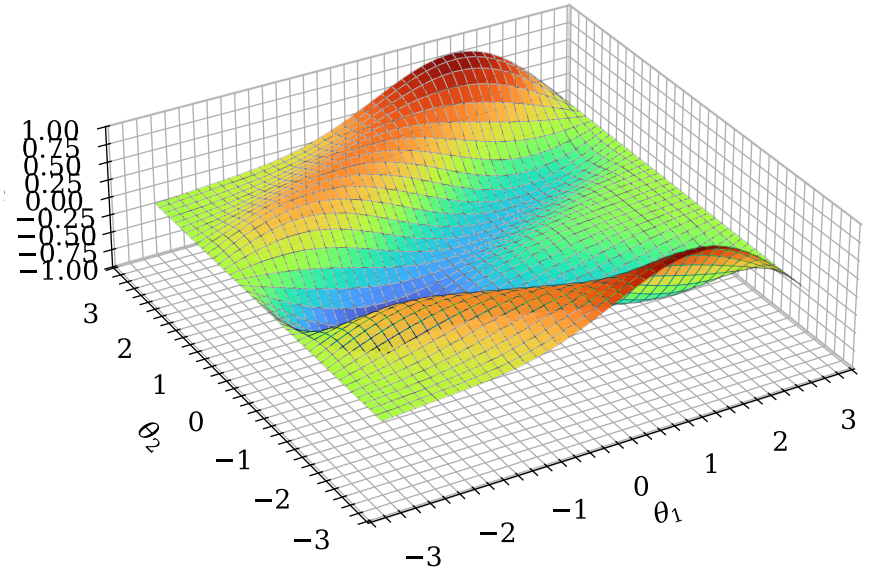
BASICS OF FIM · WI001287

Statistical Learning

0 INTRODUCTION



Univ.-Prof. Dr.-Ing. habil. Niklas Kühl
Lars Böcking

Chair for Information Systems and
Human-centric Artificial Intelligence
University of Bayreuth
TUM School of Management



Welcome

Agenda

- Lecturers of this course
- Course organization

Agenda

- Lecturers of this course
- Course organization

Hello from the team!



Univ.-Prof. Dr-Ing. habil. Niklas Kühl

Lecture/Exercise





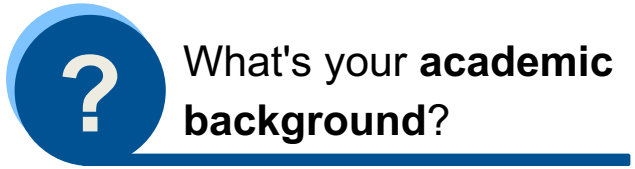
Lars Böcking

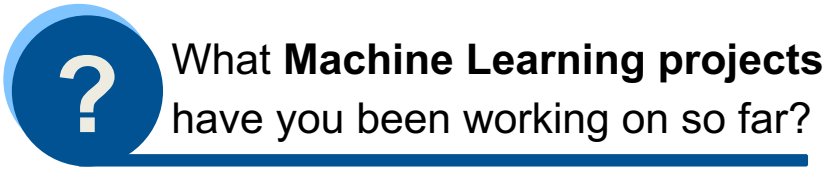
Lecture/Exercise

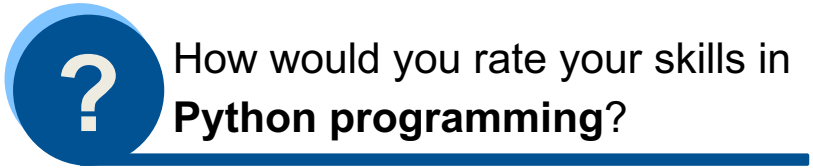


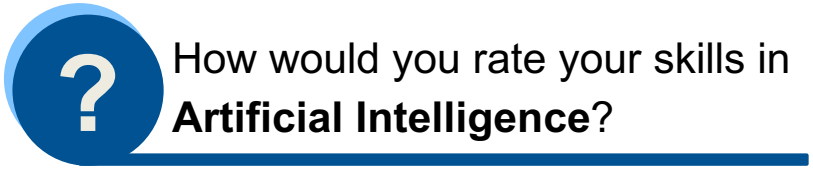
* Internship

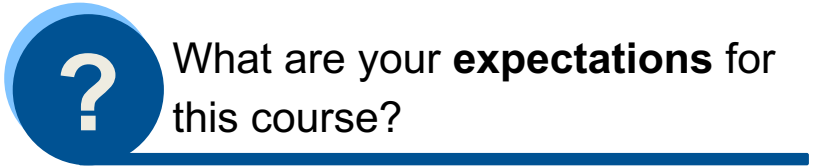
Who are you? ☺


What's your **academic background**?


What **Machine Learning projects** have you been working on so far?


How would you rate your skills in **Python programming**?


How would you rate your skills in **Artificial Intelligence**?


What are your **expectations** for this course?

Agenda

- Lecturers of this course
- Course organization

Course information and schedule

Lectures & Exercises

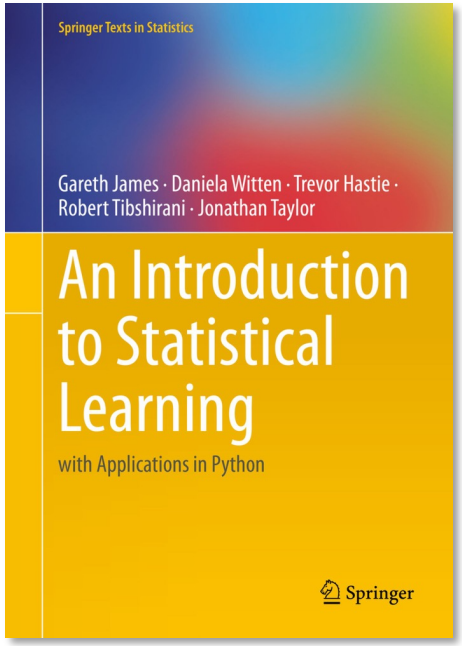
Session	Topic
1	*Fundamentals statistic & probability theory*
2	*Statistical learning and hypothesis testing*
3	*OLS regressions*
4	*Logistic regressions, Bayes‘ classification*
5	*Neural networks*

We upload all slides and Jupyter notebooks to Moodle

Exam

- ca. 1/4 of Basics of FIM final exam
- Points count towards Basics of FIM grade (6 ECTS)
- Contains both lecture content and coding questions
- No additional evaluation outside of final exam

Literature



Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, Jonathan Taylor (2023): **An Introduction to Statistical Learning**. Springer

- Course website: https://www.statlearning.com
- Book PDF: https://hastie.su.domains/ISLP/ISLP_website.pdf.download.html
- Stanford online course: https://www.edx.org/learn/python/stanford-university-statistical-learning-with-python
- Stanford online lecture recordings: https://www.youtube.com/playlist?list=PLoROMvodv4rNHU1-iPeDRH-J0cL-CrIda